

SCIENCE EXAM-2

NAME → MISHKA KHANDLWAL

CLASS → X

FORM NO. → 22254238

BATCH → BIONIRD

DATE → 14.01.2024 - SUNDAY

Answers PHYSICS

1) (3) 3:2

2) (1) ₹ 360

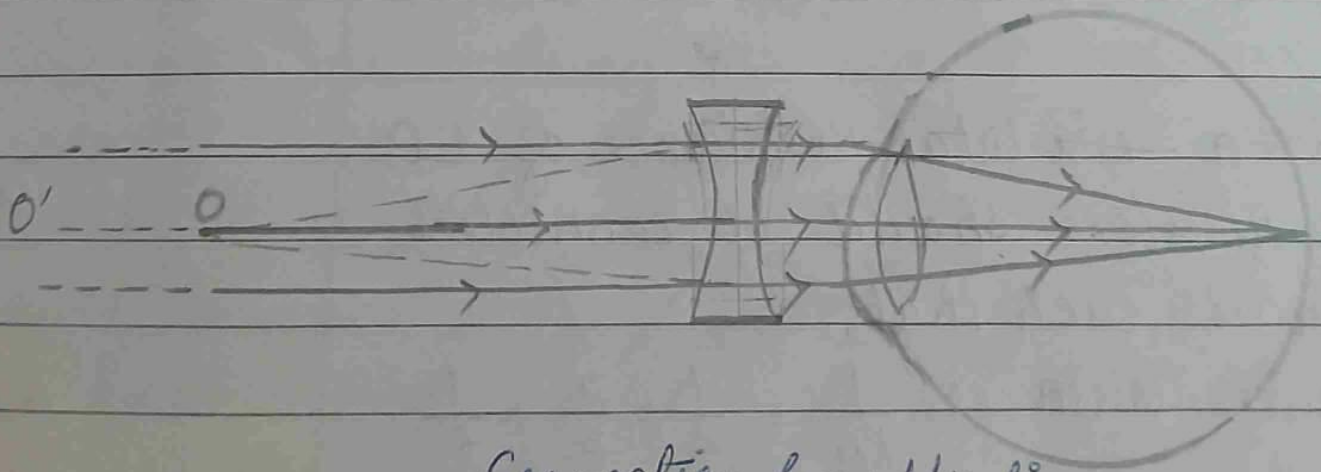
3) (3) ~~Divergent lens~~ (1) convergent lens

4) Ans → 2 common defect defects of vision :-

(1) ~~Myf~~ Myopia

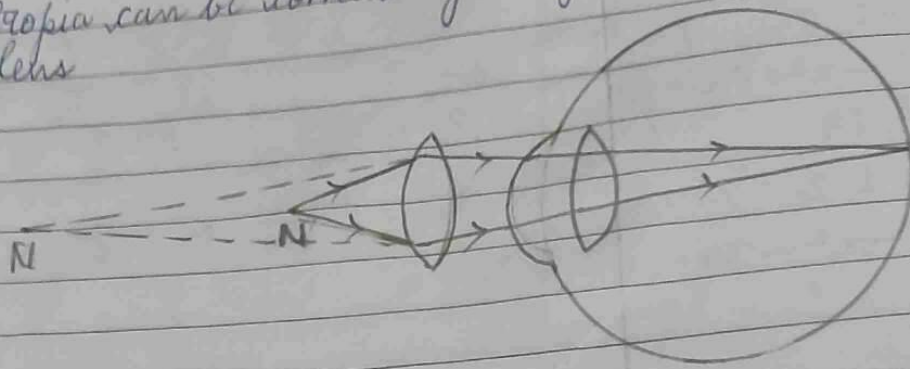
(2) Hypermetropia

Myopia can be corrected by using concave lens.



Correction for Myopia

Hypermetropia can be corrected by using
Convex lens



Correction for Hypermetropia

5)(i) Sol:- Reading of Ammeter will be the total
current flowing in the circuit.
So, now, equivalent Resistance = $8 + 5 + 12$
 $= 25 \Omega$

$$\text{Total voltage} = 2V + 2V + 2V \\ = 6V$$

$$\text{Current} = \frac{V}{R} = \frac{6}{25} = 0.24 \text{ A}$$

$\therefore$ Reading of Ammeter = 0.24 A

Reading of Voltmeter will be the voltage
across of 12Ω resistor.

$$\text{Voltage of } 12 \Omega \text{ resistor} = \frac{12}{25} \times 6 \\ = \frac{72}{25} = 2.88 \text{ V}$$

$\therefore$ Reading of Voltmeter = 2.88 V

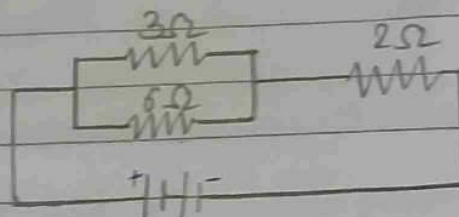
ii) Sol:- To give a total resistance of 1Ω ,
all the resistors should be connected
in parallel, such that,

$$\text{Equivalent Resistance, } \frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} \\ = \frac{1}{1} \Omega \Rightarrow R_{eq} = 1 \Omega$$

To give a total resistance of 4Ω , 3Ω and 6Ω resistors need to be connected in parallel and 2Ω resistor should be connected in series with the combination of 3Ω and 6Ω resistors, such that,

$$\frac{1}{6} + \frac{1}{3} = \frac{1+2}{6} = \frac{3}{6} \Rightarrow R_{eq} = 2\Omega$$

Now, $2\Omega + 2\Omega = 4\Omega$, which is the required total resistance.



Diagrammatic representation of 4Ω ^{total} resistance.

6) (i) ① Sol:- Far point of myopic eye, $v = -60\text{ cm}$
 $u = -\infty$

f , Focal length = ?

Using lens formula, we get,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-60} - \left(\frac{-1}{\infty}\right) = \frac{1}{f}$$

$$\Rightarrow -\frac{1}{60} = \frac{1}{f} \Rightarrow -60\text{ cm} = f$$

Now, $f = -0.6\text{ m}$

$$\text{Power, } P = \frac{1}{f} = -\frac{1}{0.6} = -\frac{10}{6} = -1.67\text{ D}$$

Nature of lens = Divergent, which is a concave lens

ii) Ans $\rightarrow$ Power of accommodation is the ability of eye to lens to adjust its focal length. This ability of eye lens is controlled by ciliary muscles.

When the ciliary muscles contract, then the focal length of eye lens is less and when focal length ciliary muscles relax, then its focal length of crystalline eye lens ~~is~~ is more.

ii) Ans → Role of iris :-

- 1) Controls the amount of light to be entered in the eye ~~on the retina~~.
- 2) Determines the colour of eye.
- 3) Controls the size of pupil.